

WOXSEN UNIVERSITY

School of Technology



PROJECT REPORT

AI-ENABLED EMERGENCY SAFETY HELMET FOR PROACTIVE WORKER HAZARD DETECTION AND LOCATION-AWARE RISK INTELLIGENCE

Submitted in partial fulfillment of the requirements for the degree of
B. Tech. in Artificial Intelligence and Machine Learning

Submitted by:

shwechith charan 2261

morla karthik 2166

keerthan-2171

Prudhvi-2124

Madhav-2081

Under the guidance of:

Santhosh kumar

Department of Artificial Intelligence and Machine Learning
Woxsen University

Hyderabad, India

October 2026

CERTIFICATE

This is to certify that the project report entitled “AI-ENABLED EMERGENCY SAFETY HELMET FOR PROACTIVE WORKER HAZARD DETECTION AND LOCATION-AWARE RISK INTELLIGENCE” submitted by [Student Name(s)] ([UGID(s)]) in partial fulfillment of the requirements for the award of the degree of B. Tech. in Artificial Intelligence and Machine Learning from Woxsen University is a bona fide record of work carried out by the students under my supervision and guidance. The work embodied in this project has been carried out as part of the academic requirements and has not been submitted elsewhere for the award of any other degree or diploma.

Signature of Mentor

Name: Santhosh kumar

Designation: [Designation]

Date: 6/10/2026

DECLARATION OF THE CANDIDATE

We hereby declare that the project work entitled “AI-ENABLED EMERGENCY SAFETY HELMET FOR PROACTIVE WORKER HAZARD DETECTION AND LOCATION-AWARE RISK INTELLIGENCE” submitted to the School of Technology, Woxsen University, in partial fulfillment of the requirements for the award of the degree of B. Tech. in Artificial Intelligence and Machine Learning is our original academic work carried out under the guidance of the faculty mentor. We further declare that the work reported in this project has not been submitted, either in part or in full, for the award of any other degree or diploma at this or any other institution.

Signature of Student

Name: shwechith charan

Roll Number: 25WU0102261

Signature of Student

Name: morla karthik 2166

Roll Number: 25WU0102166

Signature of Student

Name: keerthan

Roll Number: 25WU0102171

Signature of Student

Name: Madhav

Roll Number: 25WU0102081

Signature of Student

Name: Prudhvi

Roll Number: 25WU0102124

ACKNOWLEDGMENT

We express our sincere gratitude to our project mentor, [Mentor Name], for providing guidance, feedback, and encouragement throughout the development of this conceptual project. We thank the School of Technology and the Department of Artificial Intelligence and Machine Learning at Woxsen University for providing the academic environment and resources required for this work. We also acknowledge the researchers and engineers whose work in IoT-enabled safety systems, wearable sensing, long-range communication, and AI-based monitoring informed the development of the proposed architecture. Finally, we thank our teammates, peers, and families for their support during the project.

[shwechith charan ,morla Karthik,keerthana,Prudhvi, madhav]

ABSTRACT

This project proposes an AI-enabled emergency safety helmet for workers operating in hazardous environments, with underground mining as the primary use case. The proposed helmet integrates an ESP32 controller, MPU6050 accelerometer and gyroscope, piezoelectric impact sensing, a prototype gas sensor, LoRa communication, an emergency button, LED indicators, a buzzer, and a rechargeable power subsystem. Every helmet is assigned a unique identification number, allowing the control server to associate sensor events with individual workers and their work zones. The system detects high-impact events, suspected falls, hazardous environmental conditions, and manually triggered emergencies. Immediate events are processed at the helmet so that local warnings remain available even when network connectivity is degraded. Event packets are then transmitted to a central monitoring system for visualization and historical analysis. The principal AI contribution is a location-aware risk-intelligence layer that aggregates incidents from multiple helmets. If repeated falls, impacts, gas warnings, or emergency-button activations occur in the same zone, the system increases that zone's risk score and can classify it as a hazard hotspot. Workers subsequently approaching the hotspot can receive a preventive warning through the helmet before an incident occurs. The architecture therefore combines reactive emergency response with proactive, collective safety learning. The current work is a conceptual/prototype design and does not claim field-certified performance. Future work includes industrial-grade sensing, precise underground positioning, communication benchmarking, AI model validation, and compliance with applicable safety standards.

Keywords: Smart Helmet, Worker Safety, Internet of Things, Artificial Intelligence, Fall Detection, Gas Detection, LoRa, Risk Prediction, Hazard Hotspot, Industrial IoT

TABLE OF CONTENTS

1. INTRODUCTION

- 1.1 Background
- 1.2 Motivation
- 1.3 Problem Statement
- 1.4 Objectives
- 1.5 Scope
- 1.6 Limitations
- 1.7 Organization of the Report

2. LITERATURE REVIEW / TECHNOLOGY REVIEW

- 2.1 IoT-Enabled Safety Helmets
- 2.2 AI-Based Worker Safety Monitoring
- 2.3 Fall and Impact Detection
- 2.4 Hazardous Gas Detection
- 2.5 Long-Range Communication
- 2.6 Research Gap
- 2.7 Novelty

3. METHODOLOGY

- 3.1 System Overview
- 3.2 Hardware Components
- 3.3 Helmet Identification
- 3.4 Fall and Impact Detection
- 3.5 Emergency Button
- 3.6 Gas Hazard Detection
- 3.7 Communication Architecture
- 3.8 Location and Zone Mapping
- 3.9 AI-Based Risk Intelligence
- 3.10 Hotspot Detection and Preventive Warning
- 3.11 Server Dashboard
- 3.12 Data Processing Pipeline
- 3.13 Alert-State Logic
- 3.14 Security and Reliability Considerations

4. RESULTS AND DISCUSSION

- 4.1 Prototype/Conceptual Results

4.2 Expected Functional Outcomes

4.3 Validation Strategy

4.4 Risk-Intelligence Example

4.5 Discussion

4.6 False Alarm Management

4.7 Expected Impact

5. CONCLUSION AND FUTURE WORK

5.1 Summary of Contributions

5.2 Limitations

5.3 Future Work

5.4 Concluding Remarks

REFERENCES

APPENDIX – I: Prototype Visuals and System Flow

APPENDIX – II: Suggested Test Cases

LIST OF TABLES

Table 3.1: Proposed Hardware Components

Table 3.2: Helmet Alert States

Table 3.3: Example Event Data Schema

Table 4.1: Proposed Prototype Validation Tests

LIST OF FIGURES

Figure 3.1: Overall System Architecture

Figure 3.2: Smart Helmet Component Layout

Figure 3.3: AI Hazard-Hotspot Learning Workflow

Figure 4.1: Control Room Dashboard Concept

Figure 4.2: System Output Flow

1. INTRODUCTION

1.1 Background

Workers in underground mines, construction sites, tunnels, manufacturing plants, and other hazardous environments can be exposed to sudden impacts, falls, toxic gases, structural hazards, and communication difficulties. Conventional personal protective equipment provides essential physical protection, but a conventional helmet normally cannot detect an accident, communicate the worker's condition, or learn from repeated incidents in a particular location. Recent research has demonstrated that IoT-enabled helmets can combine sensing, communication, and emergency alerts to improve situational awareness. However, many existing approaches primarily monitor the individual worker or the immediate environment and then generate a reactive alarm.

1.2 Motivation

The motivation for this project is to move from a purely reactive safety helmet to a connected and increasingly proactive safety system. The proposed helmet is assigned a unique worker/helmet identification number and communicates sensor events to a central monitoring server. If a worker falls, experiences a high-impact event, detects a dangerous gas condition, or presses an emergency button, the event is associated with the worker ID and location/zone. The central system stores these events and analyses their spatial and temporal pattern. If repeated incidents occur in the same area, the system can classify that area as a developing risk hotspot and warn workers who subsequently approach it. The central idea is therefore collective safety intelligence: one worker's incident can help protect the next worker.

1.3 Problem Statement

Existing worker-safety helmets and monitoring systems commonly focus on individual detection, such as gas sensing, fall detection, panic buttons, or location monitoring. A key limitation is that the information from multiple workers is not always converted into a continuously updated map of local risk. Consequently, a worker may enter an area where several previous incidents or emergency activations have already occurred without receiving a contextual warning. The problem addressed by this project is the design of an AI-enabled, connected emergency helmet system that combines individual event detection, worker identification, location/zone information, real-time communication, centralized monitoring, and historical incident intelligence to support both rapid response and preventive warnings.

1.4 Objectives

The specific objectives of the project are:

1. To design a smart safety helmet capable of detecting falls, high-impact events, hazardous gas conditions, and manually triggered emergencies.
2. To assign each helmet a unique ID so that the control room can associate an event with a specific worker.
3. To transmit emergency and sensor information from helmets to a central server/dashboard using a long-range communication layer such as LoRa.
4. To provide local warnings through LEDs and a buzzer so that workers receive immediate feedback even before a control-room response arrives.
5. To develop an AI/risk-intelligence layer that aggregates historical incidents and identifies locations with repeated safety events.
6. To warn subsequent workers when they approach a location whose learned risk score exceeds a defined threshold.

7. To provide a scalable architecture in which multiple helmets contribute data to a shared safetyknowledge base.

1.5 Scope

The primary use case is underground mining and other worker environments where radio communication, environmental hazards, and delayed emergency response are important concerns. The project covers the helmet hardware, edge event detection, wireless communication, server-side monitoring, incident storage, risk scoring, and worker warning logic. The prototype is conceptual and can be extended to construction, tunnels, factories, oil and gas facilities, warehouses, and other hazardous industrial settings.

1.6 Limitations

The prototype should not be presented as a certified life-safety product. Sensor modules such as MQ-series gas sensors are suitable for prototyping but require calibration and industrial-grade sensing before deployment in safety-critical environments. Precise underground positioning cannot be assumed from LoRa alone; the proposed system therefore uses zone/gateway mapping in the conceptual design, while precise indoor positioning can be added using UWB, BLE beacons, inertial positioning, or other mine-localization infrastructure. AI warnings should support, not replace, certified safety procedures and human supervision.

1.7 Organization of the Report

Chapter 1 introduces the problem, motivation, objectives, scope, and limitations. Chapter 2 reviews related smart-helmet, IoT, AI, fall-detection, gas-monitoring, and long-range communication approaches and identifies the research gap. Chapter 3 presents the proposed system architecture and methodology. Chapter 4 discusses the prototype design, expected outcomes, validation strategy, and interpretation of the proposed system. Chapter 5 summarizes the contribution, limitations, and future work. References and appendices provide supporting technical information.

2. LITERATURE REVIEW / TECHNOLOGY REVIEW

2.1 IoT-Enabled Safety Helmets

Recent research has explored smart helmets that integrate environmental sensors, worker monitoring, wireless communication, and emergency alerts. Singh et al. presented an IoT-enabled helmet for mine-worker safety that combines environmental monitoring and location-related functionality, illustrating the value of connected personal protective equipment. Other recent smart-helmet systems use ESP-class controllers, gas sensors, accelerometers, GPS or zone information, and mobile or cloud dashboards. These systems demonstrate that the helmet can become an edge sensing node rather than only a passive protective device.

2.2 AI-Based Worker Safety Monitoring

AI can add a second layer beyond threshold-based sensing. An individual sensor can report that acceleration exceeded a threshold, but an intelligent system can combine acceleration, orientation, inactivity, emergency-button activation, historical events, and spatial patterns. Prior work has also explored AI-enabled mining helmets and machine-learning approaches for worker-state recognition. The proposed project uses AI primarily at the server level for risk accumulation and hotspot identification, while time-critical local events remain deterministic and rule-based at the edge.

2.3 Fall and Impact Detection

An inertial measurement unit such as the MPU6050 provides accelerometer and gyroscope measurements that can be used to detect sudden changes in acceleration and orientation. A robust fall detector should avoid treating every sudden movement as an emergency. The proposed approach therefore combines

multiple signals: impact magnitude, orientation change, post-event inactivity, and optionally a manual confirmation window. A piezoelectric impact sensor can provide an additional impact signal. Sensor fusion reduces the probability that a single noisy reading creates a false emergency.

2.4 Hazardous Gas Detection

Gas monitoring is a major component of mine-worker safety systems. MQ-series sensors are commonly used in educational and prototype systems, including recent smart-helmet projects. In the proposed architecture, the gas sensor provides a local hazard indicator that can trigger a warning and send the event to the server. For deployment, the sensor should be replaced or supplemented with calibrated industrial gas detectors appropriate to the target environment and specific gases of concern, especially methane and carbon monoxide in coal-mining applications.

2.5 Long-Range Communication

LoRa/LoRaWAN-type communication is attractive for industrial sensing because it is designed for long-range, low-power transmission of small, intermittent sensor messages. This fits a helmet system in which the important messages are compact event packets rather than continuous high-bandwidth media. The architecture can use LoRa nodes on helmets and gateways positioned around the worksite, with a server receiving and processing event data.

2.6 Research Gap

The reviewed systems establish that smart helmets can detect hazards, transmit alerts, and support remote monitoring. The central gap addressed by this project is the transition from individual reactive monitoring to collective, location-aware risk intelligence. Instead of treating each event as an isolated alert, the proposed system stores events from many helmet IDs and analyses recurrence by location and time. A repeated pattern can increase the risk score of a zone, leading to a preventive warning for another worker before an accident occurs. This creates a feedback loop: detection generates data; data generates a risk model; the risk model generates preventive warnings; new observations then update the model.

2.7 Novelty

The main novelty is not the presence of a single sensor or communication module. The contribution is the integration of (1) unique helmet identity, (2) multi-sensor emergency detection, (3) real-time location/zone association, (4) centralized event history, and (5) AI-driven spatial risk learning. The system is designed so that a hazard discovered by one worker can improve the safety of other workers. This changes the helmet from an isolated emergency detector into a node in a collective safety intelligence network.

3. METHODOLOGY

3.1 System Overview

The proposed system has five major layers: sensing, edge processing, wireless communication, server intelligence, and user alerts. Each helmet acts as a wearable IoT node. The ESP32 receives sensor readings and monitors emergency conditions. When an event is detected, the helmet creates an event packet containing the helmet ID, event type, sensor values, timestamp, and location/zone information. The packet is transmitted through the LoRa communication layer to a gateway and then to the central server. The server updates the worker state, event history, live dashboard, and location risk model.

3.2 Hardware Components

The conceptual prototype uses an ESP32 microcontroller as the main controller; an MPU6050 accelerometer/gyroscope for motion and orientation sensing; a piezoelectric impact sensor for strong impact detection; an MQ-135-type gas sensor for prototype environmental sensing; a LoRa SX1278 communication module for long-range wireless communication; a high-power LED headlamp; a red

status LED; a manual emergency button; a piezo buzzer; a rechargeable 3.7 V 18650 lithium-ion battery; and a TP4056 charging module. The helmet also requires protective mounting, internal wiring, insulation, ventilation around the gas sensor, and comfortable suspension/padding.

3.3 Helmet Identification

Every helmet is assigned a unique identifier such as HLM-001, HLM-002, or a worker-specific ID. The ID is included in every event packet. This allows the server to distinguish between multiple workers and maintain individual histories. The system can associate the helmet ID with a worker profile, assigned work zone, shift, and emergency contact information. In a production system, identity management should also support secure authentication so that unauthorized devices cannot inject false safety events.

3.4 Fall and Impact Detection

The edge controller continuously samples accelerometer and gyroscope data. A simplified detection sequence is: (1) identify a sudden acceleration spike or abnormal motion; (2) examine orientation change; (3) check whether the worker becomes inactive after the event; (4) cross-check the piezoelectric impact sensor where available; and (5) classify the event as normal movement, suspected fall, or high-impact emergency. The system can provide a short local cancellation interval for false alarms, but a severe impact or combined evidence can immediately trigger the emergency state. This logic is intentionally placed at the edge because emergency detection should not depend entirely on cloud connectivity.

3.5 Emergency Button

A dedicated physical emergency button allows the worker to report danger even when automatic sensors cannot identify the event. Pressing the button generates a high-priority emergency packet containing the worker/helmet ID, timestamp, and location/zone. The helmet activates a red LED and buzzer, while the server marks the worker as requiring assistance. The button is important because not every dangerous situation produces a measurable fall or impact; for example, a worker may encounter a blocked passage, fire, medical problem, threatening environmental condition, or another hazard.

3.6 Gas Hazard Detection

The gas sensor continuously provides an environmental signal. At the prototype level, threshold values can be configured for warning and critical states after calibration. A warning state can trigger a yellow indicator and slower buzzer pattern, while a critical state can trigger a red indicator, continuous buzzer, and immediate server alert. The server stores gas-related events by location, allowing repeated gas alarms in one zone to contribute to the zone risk score. Production deployment requires calibrated sensors and safety-certified threshold definitions rather than arbitrary prototype thresholds.

3.7 Communication Architecture

The helmet transmits compact packets through the LoRa module to a nearby gateway. The gateway forwards data to the server using an available backhaul such as Ethernet, Wi-Fi, cellular, or another mine network. The server exposes a dashboard showing active workers, emergency alerts, historical incidents, zone risk, battery status, and communication status. The design is suitable for intermittent telemetry because the system does not require continuous video or large data transfers.

3.8 Location and Zone Mapping

Because underground GNSS/GPS availability can be poor or absent, the proposed conceptual system treats location as a mine-defined zone or tunnel/section identifier. LoRa gateways, fixed beacons, RFID checkpoints, UWB anchors, BLE beacons, or other localization mechanisms can map a helmet to a zone. The server stores the location associated with every event. For example, repeated emergency events can be grouped under “Section B – Tunnel 4” rather than relying on a consumer GPS receiver. Precise positioning is a future hardware enhancement if required.

3.9 AI-Based Risk Intelligence

The central AI feature is a location-risk model. Each incident is converted into a structured record: helmet ID, location, time, event type, severity, sensor values, and response status. Events are aggregated by spatial zone and time window. A risk score can be conceptually defined as $R(z,t) = w_1E + w_2F + w_3G + w_4M + w_5Q$, where E represents emergency-button activations, F represents fall/impact events, G represents gas warnings, M represents severity-weighted incidents, Q represents recent-event density, and the w terms are learned or configured weights. The score should also decay over time so that an old incident does not permanently label a location as dangerous.

3.10 Hotspot Detection and Preventive Warning

When the risk score of a zone crosses a defined warning level, the server marks the zone as a risk hotspot. If another worker's helmet is detected in or approaching that zone, the server sends a contextual warning. The helmet can display the warning through a yellow LED and/or buzzer pattern, while the control dashboard can show the reason for the warning, such as repeated fall events or repeated emergency activations. This is the preventive loop of the project: previous incidents create knowledge that changes the behavior of future workers.

3.11 Server Dashboard

The control-room dashboard provides a consolidated view of worker IDs, live status, emergency events, location/zone, battery level, signal status, gas state, incident history, and risk hotspots. An emergency panel can rank incidents by severity. A map or mine layout can highlight the worker and the affected zone. The dashboard also provides historical analytics so supervisors can identify recurring hazards and prioritize corrective action such as ventilation, structural inspection, route closure, additional signage, or worker reassignment.

3.12 Data Processing Pipeline

The complete data pipeline is: sensor acquisition → edge filtering → event detection → helmet ID attachment → timestamp and location tagging → LoRa transmission → gateway → server ingestion → database storage → AI risk analysis → dashboard update → worker warning/rescue notification. This architecture separates safety-critical immediate detection from higher-level AI analytics. If the network is temporarily unavailable, the helmet should continue local detection and warning, and the system can buffer important events for later transmission where memory permits.

3.13 Alert-State Logic

The proposed states are NORMAL, WARNING, and EMERGENCY. NORMAL indicates that monitored values are within configured limits. WARNING indicates a developing hazard such as elevated gas reading or entry into a high-risk zone. EMERGENCY indicates a severe impact, confirmed fall, or manual emergency activation. Each state has a local helmet indication and a corresponding server action. This makes the interface understandable for workers operating in noisy or visually demanding environments.

3.14 Security and Reliability Considerations

Worker safety data should be protected because it includes identity, movement, and incident history. Communication should use device authentication and encryption where supported, and the server should enforce role-based access. The system should log all emergency events and changes to risk classifications. Hardware should be mechanically protected, water-resistant, vibration-resistant, and designed so that wiring does not interfere with the helmet's certified protective structure. Battery monitoring and low-power operation are also necessary for practical deployment.

4. RESULTS AND DISCUSSION

4.1 Prototype/Conceptual Results

The current work establishes a complete conceptual architecture and product-level design for an AI-enabled emergency helmet rather than claiming completed field validation. The supplied design visualizations show a helmet with an ESP32 controller, MPU6050 motion sensing, piezoelectric impact sensing, MQ-135-type gas sensing, LoRa communication, LED headlamp, status LED, battery/charging subsystem, buzzer, and emergency-alert interface. The associated dashboard concept shows worker identification, emergency state, impact value, gas status, battery, signal strength, location/section, time, rescue status, scenario-specific alerts, mobile notification, live map view, and an output flow from event detection to rescue initiation.

4.2 Expected Functional Outcomes

The system is expected to achieve the following functional outcomes during controlled prototype testing:

- Detect a simulated high-impact event and generate a local emergency indication.
- Detect a simulated fall pattern using inertial measurements and inactivity confirmation.
- Allow a worker to manually trigger an emergency using the physical button.
- Detect elevated prototype gas-sensor readings and generate a warning state.
- Associate each event with the correct helmet ID.
- Transmit compact event packets through the LoRa communication path.
- Display active events and worker state on the control dashboard.
- Accumulate repeated incidents by zone and calculate an updated risk score.
- Generate a preventive warning when a worker approaches a high-risk zone.

4.3 Validation Strategy

Validation should be performed as a series of controlled tests rather than by inventing accuracy figures. Fall detection can be tested using safe simulated movements and measured inertial patterns. Impact detection can be tested using controlled, non-destructive sensor stimulation. The emergency button can be tested repeatedly for packet reliability and server acknowledgement. Gas sensing should be tested only with safe, controlled calibration procedures and appropriate laboratory supervision. Communication testing should record packet delivery rate, latency, range within the intended environment, and behavior when connectivity is temporarily lost. AI hotspot testing can use a synthetic event dataset in which multiple incidents are deliberately assigned to the same zone and the system is checked for correct risk-score escalation and warning generation.

4.4 Risk-Intelligence Example

Consider a mine divided into zones A, B, C, and D. If three different helmets report falls or emergency activations in Zone C within a recent time window, the server should not treat the reports as unrelated. Their combined evidence increases the risk score of Zone C. A supervisor can then inspect the area. Until the risk is resolved or the score decays, another worker entering Zone C can receive a warning. If subsequent data show no further incidents after corrective action, the risk score can gradually decrease. This example illustrates why the proposed system is more than a simple alarm: it creates a continuously updated safety memory.

4.5 Discussion

The architecture provides several advantages. First, emergency detection is distributed to the worker, reducing dependence on a worker noticing and reporting an incident manually. Second, communication allows supervisors to see the event without waiting for the worker to reach a safe location. Third, multi-worker data creates a broader picture of site safety. Fourth, the AI layer can identify patterns that are difficult to see from isolated alerts. Finally, the design is scalable because additional helmets can be registered without changing the fundamental event model.

4.6 False Alarm Management

False alarms are a major concern in safety systems. A helmet must not interpret every sudden movement as a fall. The proposed sensor-fusion approach reduces this risk by combining acceleration, orientation, impact evidence, and post-event movement. Likewise, a single noisy gas-sensor reading should not automatically classify a zone as permanently hazardous. The AI layer should use repeated evidence, severity, recency, and confidence. Human supervisors should also be able to acknowledge, investigate, and clear a hotspot after corrective action.

4.7 Expected Impact

The expected impact is a shift from reactive incident notification toward proactive risk prevention. Traditional emergency notification answers the question “Who is in trouble now?” The proposed system additionally asks “Where are incidents repeatedly happening, why might that area be risky, and who should be warned before entering?” This distinction is the central value proposition of the project. It can support faster emergency response while also helping supervisors identify recurring hazards and improve site-level safety practices.

5. CONCLUSION AND FUTURE WORK

5.1 Summary of Contributions

This project proposes an AI-enabled emergency safety helmet in which every helmet becomes a uniquely identified node in a connected worker-safety network. The helmet combines motion and impact sensing, gas monitoring, a manual emergency button, local visual/audible alerts, long-range communication, and server-side monitoring. The major contribution is the collective risk-intelligence layer: incident data from multiple workers are associated with location and time, allowing the system to learn recurring risk hotspots and warn future workers. The architecture therefore combines detection, communication, response, and prevention in a single safety ecosystem.

5.2 Limitations

The current design is a conceptual/prototype architecture and requires extensive physical validation before safety-critical deployment. Prototype gas sensors require calibration and may not provide the selectivity or certification needed for hazardous mines. Underground localization requires dedicated infrastructure if precise positioning is required. LoRa communication performance depends on mine geometry, antenna placement, gateway locations, interference, and regulatory constraints. AI risk scoring also requires representative historical data and careful tuning to avoid excessive false warnings. Finally, the electronics must be integrated without compromising the mechanical and regulatory certification of the underlying safety helmet.

5.3 Future Work

Future development can include industrial-grade methane and carbon-monoxide sensors, certified impact sensors, UWB/BLE-based indoor positioning, additional environmental sensors, worker physiological monitoring where ethically and technically justified, edge machine learning for personalized fall detection, predictive maintenance of mine infrastructure, digital-twin visualization of mine risk, and automated integration with emergency response systems. The AI model can be expanded from hotspot detection to predictive risk forecasting using time-series data. Federated or privacy-preserving learning could also be investigated when multiple worksites contribute data without sharing raw worker records.

5.4 Concluding Remarks

The proposed AI-enabled emergency helmet demonstrates how wearable IoT systems can evolve from simple alarm devices into intelligent safety networks. Its most important concept is collective learning: a dangerous event reported by one worker becomes information that can protect another worker. By combining immediate edge-level detection with server-side historical analysis and location-aware risk intelligence, the system has the potential to support faster emergency response and proactive hazard prevention. The next stage should focus on controlled prototype testing, sensor calibration, communication benchmarking, risk-model validation, and compliance with applicable occupational safety standards before any real-world safety-critical use.

TABLES

Table 3.1: Proposed Hardware Components

Component	Purpose	Role in System
ESP32	Main controller	Sensor acquisition, local logic, packet generation
MPU6050	Accelerometer + gyroscope	Fall, motion and orientation detection
Piezoelectric impact sensor	Impact sensing	Additional evidence for strong impacts
MQ-135-type sensor	Prototype gas sensing	Environmental hazard indication
LoRa SX1278	Long-range communication	Helmet-to-gateway telemetry
Emergency button	Manual emergency trigger	Worker-initiated SOS
LED + buzzer	Local alerts	Visual/audible worker warning
18650 Li-ion battery	Power	Portable energy source
TP4056	Battery charging	Rechargeable battery management

Table 3.2: Helmet Alert States

Condition	LED	Buzzer	Server Action
Normal	Green/normal status	Off	Routine monitoring
Zone Warning / Gas Warning	Yellow or warning indicator	Slow/intermittent beep	Store event; warn worker
High Impact / Confirmed Fall	Red blinking	Continuous beep	Emergency alert + rescue workflow
Manual Emergency	Red blinking	Continuous beep	Immediate high-priority alert

Table 3.3: Example Event Data Schema

Field	Example	Purpose
helmet_id	HLM-021	Unique device/worker identifier
event_type	FALL / IMPACT / GAS / SOS	Classifies safety event
timestamp	2026-10-06 00:10:30	Event chronology
zone_id	SECTION-B-TUNNEL-4	Location association
severity	HIGH	Priority
sensor_values	accel, gyro, gas	Evidence for event
battery	82%	Device health
signal	Strong	Communication health

Table 4.1: Proposed Prototype Validation Tests

Test	Input	Expected Output	Metric
Fall detection	Safe simulated fall motion	Fall/emergency state	Detection rate, false alarms
Impact detection	Controlled sensor impact	High-impact alert	Trigger consistency
SOS button	Manual press	Immediate emergency packet	Packet latency/success
Gas warning	Controlled calibrated test	Warning/critical state	Response time
LoRa link	Event packet	Gateway receives packet	Delivery rate, latency
Hotspot AI	Repeated events in one zone	Risk score increases	Correct hotspot classification
Preventive warning	Worker enters hotspot	Helmet warning	Warning delivery rate

REFERENCES

- [1] N. Singh, V. K. Gunjan, G. Chaudhary, R. Kaluri, N. Victor, and K. Lakshmanan, "IoT enabled HELMET to safeguard the health of mine workers," *Computer Communications*, vol. 193, 2022, doi: 10.1016/j.comcom.2022.06.032.
- [2] S. Karna, T. Noushin, and S. Tabassum, "IoT based Smart Helmet for Automated and Multi-parametric Monitoring of Underground Miners' Health Hazards," *2022 IEEE 15th Dallas Circuit and System Conference (DCAS)*, 2022, doi: 10.1109/DCAS53974.2022.9845621.
- [3] "AI-Based Safety Helmet for Mining Workers Using IoT Technology and ARM Cortex-M," *IEEE Sensors Journal*, vol. 23, no. 18, pp. 21355–21362, 2023, doi: 10.1109/JSEN.2023.3296523.
- [4] M. Mariappan, "Design and Implementation of Long Range Wide Area Networks for Future Industrial IoT Applications," *International Journal of Sensors, Wireless Communications and Control*, vol. 14, no. 3, pp. 215–225, 2024.
- [5] Espressif Systems, "ESP32 Series Datasheet," Espressif Systems, latest available version.
- [6] K. S. Rao, S. Naveen Kumar, D. Shreya Sree, P. Priyadarshini, and A. Sandeep Kumar, "IoT Based Multi-Functional Smart Helmet for Safety Monitoring and Alerting of Mine Workers," *International Journal of Creative Research Thoughts*, vol. 12, no. 2, 2024.
- [7] M. Ali, P. Kumari, A. Vishwakarma, A. Madankar, and A. Pahadia, "IoT Based Coal Mine Safety

Monitoring and Alerting Helmet System,” *International Journal of Research & Technology*, vol. 14, no. 2, 2026, pp. 951–956.

[8] K. Bidkar, K. Rathod, R. Shaikh, R. Badki, S. Meche, and H. Sahare, “Smart Helmet for Mine Workers,” *International Journal of Data Science and IoT Management System*, vol. 5, no. 1, 2026, pp. 1–4.

[9] A. N. Reddy, B. Vishnu Teja, D. Gupta, and G. Madhavi, “Coal Mine Helmet Safety Monitoring System Based on IoT,” *International Research Journal of Modernization in Engineering Technology and Science*, vol. 7, no. 5, 2025.

[10] M. E. Nilsback and A. Zisserman, “Automated flower classification over a large number of classes,” in *Proc. 2008 Sixth Indian Conference on Computer Vision, Graphics & Image Processing*, 2008, pp. 722–729. (Included as a methodological reference for supervised classification concepts; not a direct mining-safety source.)

APPENDIX – I

Prototype Visuals and System Flow

Insert the supplied smart-helmet product visualization, multi-view engineering visualization, and control-room dashboard visualization in this appendix. The figures should be captioned according to Woxsen guidelines. The supplied design depicts the helmet hardware, internal component arrangement, alert states, live-map concept, and the event-to-control-room flow.



APPENDIX – II

Suggested Prototype Test Cases

TC-01: Register a helmet and verify that its unique ID appears correctly on the server.

TC-02: Trigger the emergency button and verify local LED/buzzer response, packet transmission, and dashboard alert.

TC-03: Simulate a fall pattern and verify sensor-fusion classification.

TC-04: Trigger a controlled impact-sensor test and verify emergency escalation.

TC-05: Generate a controlled gas-sensor warning and verify local and remote alerts.

TC-06: Generate repeated incidents from different helmet IDs in the same zone and verify risk-score escalation.

TC-07: Move another helmet into the high-risk zone and verify that a preventive warning is generated.

TC-08: Disconnect the network and verify that local emergency detection remains operational.

TC-09: Restore communication and verify buffered critical events are synchronized where supported.

TC-10: Verify that authorized dashboard users can view incidents and that event history is retained.